



GREEN HYDROGEN DEMAND AND REFUELLING NETWORK FOR HEAVY-DUTY TRUCKING IN INDIA

June 2026



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Technical Annex



Abbreviations, Symbols and Units

C: Truck count at toll plaza

Capacity: Cumulative national capacity of HRS

D: Fuel demand

$d_{i-1,i}$: Distance to upstream plaza

$d_{i,i+1}$: Distance to downstream plaza

FC: Fuel consumption

FE_{diesel} : Average fuel economy (km/L)

HCV: Heavy commercial vehicles

HRS: Hydrogen refuelling station

L: Effective road segment/ link length represented by plaza i

LHV: Lower heating value of the fuel

N: Number of HRS of capacity x tpd

n: Number of toll plazas

NH: National highway

R: Replacement ratio

Res: Residual demand

VKT: Vehicle kilometres travelled

η_{FCEV} : Efficiency of fuel cell EV

η_{ICE} : Efficiency of IC engine

η_{FC} : Fuel cell efficiency

η_{PT} : Powertrain efficiency

δ : Binary indicator variable accounting for residual demand

Δ : Overcapacity

Subscripts

avg: Average value/best estimate

best: Best value estimate

FCEV: Fuel cell electric vehicles

FC: Fuel cell

diesel: Diesel fuel

hydrogen, H_2 : Hydrogen fuel

high: Higher bound

i: Toll Plaza notation

low: Lower bound

N: Toll plaza notation

PT: Powertrain

1, 2, 3, 4, 5: HRS Capacity, tonnes per day

Superscripts

daily: Daily average

annual: Annual value

total: Cumulative national value

installed: Installed capacity of HRS

0.75, 0.5, 0.25, 0.1: Replacement scenarios

Units

GW: Gigawatt

kg: Kilogram

km: Kilometre

L: Litre

km/L: Kilometres per litre

MMT: Million metric tonnes

MJ/kg: Megajoules per kg

Mt: Million tonnes

TPD: Tonnes per day

A.1 Methodology

A.1.1 Estimation of Vehicle Kilometres Travelled (VKT)

Vehicle kilometres travelled (VKT) for medium and heavy-duty trucks are estimated using a traffic-count-based approach. In the absence of origin-destination level freight data, VKT is derived by combining observed traffic volumes with representative road segment lengths, thereby quantifying total vehicle movement on the network rather than individual trip distances^{5,6}.

Our analysis uses annual traffic counts at toll plazas across the Indian national highway network, segregated based on different vehicle categories such as cars, buses, and commercial vehicles. For Indian national highways, the vehicle counts at every toll plaza are published on a monthly basis for different vehicle categories by the Indian Highways Management Company Limited (IHMCL)⁷.

For our estimates, we have considered 2025 as the baseline year. The dataset comprises over 1,100 toll locations, representing a near-complete coverage of major tolled highway corridors in India. The toll plaza details are related to their respective national highway (NH) number, state, district and section of highway by comparing it with the “NH Fee Plaza” database maintained by Indian Highways Management Company⁸. Individual NH is mapped separately to determine the locations and sequence of these toll plazas by using their coordinates as a reference. The effective link length associated with each plaza was calculated as half the distance to the upstream plaza plus half the distance to the downstream plaza.

Total VKT is computed as:

$$VKT = \sum_{i=1}^n (C_i^{annual} \times L_i)$$

where:

- C_i^{annual} : annual truck count at toll plaza i
- L_i : effective road segment/link length represented by plaza i
- n : total number of toll plazas

Definition of link length: The effective segment length L_i for each toll plaza, is defined using a midpoint-to-midpoint marking approach. Each segment is also assigned a unique segment_id during analysis and is also defined by the toll plaza, which acts as the pivot/node for identification.

For an interior toll plaza i :

$$L_i = \frac{d_{i-1,i}}{2} + \frac{d_{i,i+1}}{2}$$

Where:

- $d_{i-1,i}$ = distance to the downstream plaza
- $d_{i,i+1}$ = distance to the upstream plaza

For boundary plazas:

$$L_1 = \frac{d_{1,2}}{2}, L_N = \frac{d_{N-1,N}}{2}$$

This method ensures that each kilometre of roadway is counted exactly once and there is no overlap or omission occurring across adjacent segments. Unlike conventional traffic surveys, the present dataset consists of annual traffic counts and comprehensive spatial coverage of toll plazas. Therefore, temporal expansion factors (e.g., daily-to-annual scaling) and spatial expansion factors (e.g., sample-to-network scaling) are not required. This simplifies the estimation and reduces uncertainty compared to sampled traffic-count approaches.

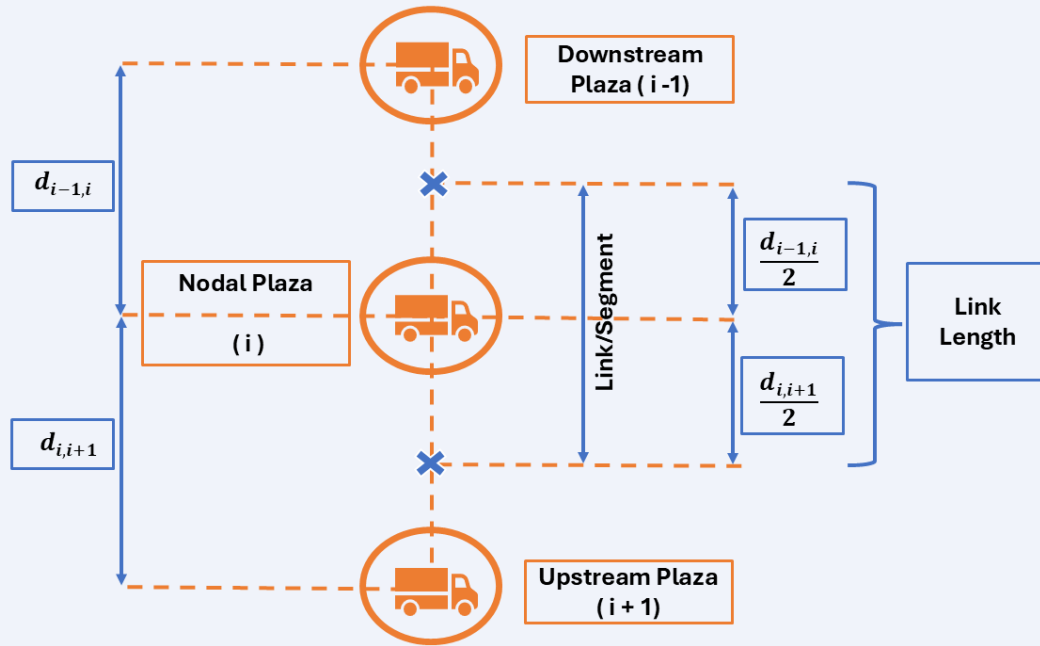
Traffic counts are listed separately for different classes of vehicles, such as car-jeep count, LCV count, bus-truck count, 3-axle count, 4_6-axle counts and other small vehicle counts. For our analysis, we have considered the 3-axle and 4_6-axle counts, which are representative categories for HCVs in India.

Key Assumptions

- The data considers 2025 traffic counts as the baseline year data.
- It is assumed that traffic observed at each toll plaza is representative of vehicle movement over the corresponding road segment. Hence, point observations are used to infer link-level activity.
- Each inter-plaza segment is assumed to exhibit relatively uniform traffic conditions. While traffic may vary due to intermediate entries/exits, such variations are expected to average out across the large number of segments considered.
- Traffic counts at different toll plazas are treated as independent measurements of segment usage, rather than unique vehicle trips. Consequently, the same vehicle may be counted multiple times across different segments, which is appropriate since VKT measures total road usage.
- Annual counts already include seasonal variation

- The toll plaza dataset is assumed to capture the majority of freight movement on major national highways. Non-tolled roads and minor freight routes are not explicitly included, and therefore, the estimates may represent a conservative approximation of total national truck activity.

Figure 1: Definition of a segment/link length



A.1.2 Estimation of Diesel Consumption

To accurately model the hydrogen-based transition of the HCV sector, establishing an energy equivalency metric between diesel internal combustion engines (ICE) and hydrogen fuel cell electric vehicles (FCEVs) is critical. Here, we derive a model based on tractive work requirement (useful work at the wheel). This approach ensures that diesel and hydrogen fuels are compared on the basis of useful mechanical work delivered at the wheels, rather than solely on fuel energy content.

Total diesel demand is calculated as:

$$D_{diesel} = \frac{VKT_{total}}{FE_{diesel}}$$

where:

- D_{diesel} : annual diesel consumption (L),
- FE_{diesel} : average fuel economy (km/L).

Indian HCVs operating in categories such as multi-axle rigid, heavy tipper, tractor-trailer, cement/bulk haulage, etc., operate on different terrains such as long-haul highways, flat roads, mining/off-road, hills, or mixed terrains. National think tanks like NITI Aayog have reported a representative fuel consumption of HGVs in India to be around 39 litres per 100 km^{2,9}. However, to account for the uncertainty, variations in payload, terrain, congestion, and vehicle condition should be reflected. Hence, the fuel economy for Indian HCVs is assumed to vary within 2.25-4 km/L as tested in real-world scenarios and reported by manufacturers. A fuel economy of 2.5 km/L is assumed as the best estimate value.

A.1.3 Hydrogen Demand Estimation

Hydrogen demand is derived by equating the useful tractive energy delivered by the diesel ICE and hydrogen FCEV.

$$FC_{diesel} \cdot LHV_{diesel} \cdot \eta_{ICE} = FC_{H_2} \cdot LHV_{H_2} \cdot \eta_{FCEV}$$

To explicitly capture drivetrain performance, the fuel cell electric vehicle (FCEV) efficiency is decomposed as:

$$\eta_{FCEV} = \eta_{FC} \times \eta_{PT}$$

Thus, the diesel-to-hydrogen replacement ratio becomes:

$$R = \frac{LHV_{H_2} \cdot \eta_{FC} \cdot \eta_{PT}}{LHV_{diesel} \cdot \eta_{ICE}}$$

Substituting into the hydrogen demand equation:

$$D_{hydrogen} = \frac{VKT_{total}}{FE_{diesel}} \cdot \frac{LHV_{diesel} \cdot \eta_{ICE}}{LHV_{H_2} \cdot \eta_{FC} \cdot \eta_{PT}}$$

Parameter Uncertainty

The transition from diesel to hydrogen-based freight transport involves multiple sources of uncertainty arising from vehicle performance, operational variability, and technology maturity. To ensure robustness of the results, an uncertainty treatment was implemented combining parametric sensitivity analysis and scenario-based bounds. Uncertainty is introduced through key performance parameters:

Table 1: Uncertainty bounds

Parameter	Central/Best Estimate Value	Range	Source of Uncertainty
Diesel ICE efficiency (η_{ICE})	0.35	0.30-0.40	Variation across load cycles and engine age
Fuel cell system efficiency (η_{FC})	0.50	0.45-0.60	Stack performance and auxiliary loads
HCV fuel economy (FE_{diesel} , km/L)	2.5	2.25-4	Payload, terrain, congestion
Powertrain efficiency, (η_{PT})	0.925	0.90-0.95	Vehicle design and duty cycle

Lower heating values are assumed constant:

$$LHV_{diesel} = 42.5 \text{ MJ/kg or } 35.8 \text{ MJ/L}, LHV_{H_2} = 120 \text{ MJ/kg}$$

Hydrogen demand, hence, becomes a function of four independent variables:

$$D_{hydrogen} = f(FE_{diesel}, \eta_{ICE}, \eta_{FC}, \eta_{PT})$$

Uncertainty is evaluated by defining the minimum (optimistic case) and maximum (conservative case) hydrogen demand as:

Optimistic Case (Higher fuel efficiency of HCV, higher FCEV efficiency):

$$D_{hydrogen,low} = \frac{VKT_{total}}{FE_{high}} \cdot \frac{LHV_{diesel} \cdot \eta_{ICE,low}}{LHV_{H_2} \cdot \eta_{FC,high} \cdot \eta_{PT,high}}$$

Conservative Case (Lower fuel efficiency of HCV, lower FCEV efficiency):

$$D_{hydrogen,high} = \frac{VKT_{total}}{FE_{low}} \cdot \frac{LHV_{diesel} \cdot \eta_{ICE,high}}{LHV_{H_2} \cdot \eta_{FC,low} \cdot \eta_{PT,low}}$$

Best Estimate:

$$D_{hydrogen,best} = \frac{VKT_{total}}{FE_{best}} \cdot \frac{LHV_{diesel} \cdot \eta_{ICE,avg}}{LHV_{H_2} \cdot \eta_{FC,avg} \cdot \eta_{PT,avg}}$$

By holding transport activity (or VKT) constant and resolving fuel substitution through efficiency-based energy equivalence, this methodology provides a transparent estimation of diesel and hydrogen demand. The uncertainty bounds capture variability in real-world vehicle performance.

A.1.4 Hydrogen Refuelling Station Allocation Methodology

To translate hydrogen demand into refuelling infrastructure requirements, a segment-capacity-based allocation model was developed. The approach determines the number and size distribution of hydrogen refuelling stations (HRS) required to meet spatially distributed daily hydrogen demand along freight corridors. For each corridor segment i , the daily hydrogen demand is derived from annual hydrogen consumption. HRS infrastructure is modelled using standardised station capacities reflecting the current status globally. Five station sizes are considered:

$$C \in \{1,2,3,4,5\} \text{ tonnes/day}$$

Capacity Allocation Logic

For each segment i , the total daily hydrogen demand is decomposed into a combination of discrete station capacities.

Step 1: Allocation of 5 TPD Stations

The number of 5 TPD stations is determined by using a floor operator on the daily segment demand:

$$N_{5,i} = \left\lfloor \frac{D_{hydrogen,i}^{daily}}{5} \right\rfloor + \delta_5$$

where:

$$\delta_5 = \begin{cases} 1, & \text{if } \text{mod}(D_{hydrogen,i}^{daily}, 5) > 4 \\ 0, & \text{otherwise} \end{cases}$$

Step 2: Residual Demand

The remaining demand after allocating 5 TPD stations is:

$$Res_i = \text{mod}(D_{hydrogen,i}^{daily}, 5)$$

Step 3: Allocation of Smaller Capacity Stations

The residual demand Res_i is met using a single station of appropriate size, based on the following conditions:

$$N_{4,i} = \begin{cases} 1, & 3 < Res_i \leq 4 \\ 0, & \text{otherwise} \end{cases}$$

$$\begin{aligned}
N_{3,i} &= \begin{cases} 1, & 2 < Res_i \leq 3 \\ 0, & \text{otherwise} \end{cases} \\
N_{2,i} &= \begin{cases} 1, & 1 < Res_i \leq 2 \\ 0, & \text{otherwise} \end{cases} \\
N_{1,i} &= \begin{cases} 1, & 0 < Res_i \leq 1 \\ 0, & \text{otherwise} \end{cases}
\end{aligned}$$

Step 4: Total Station Count

The total number of HRS units for segment i is:

$$N_i^{total} = N_{5,i} + N_{4,i} + N_{3,i} + N_{2,i} + N_{1,i}$$

Step 5: Aggregate Infrastructure Requirement

At the national or corridor level:

$$\begin{aligned}
N^{total} &= \sum_i N_i^{total} \\
Capacity^{total} &= \sum_i (5N_{5,i} + 4N_{4,i} + 3N_{3,i} + 2N_{2,i} + N_{1,i})
\end{aligned}$$

This ensures that installed capacity meets or slightly exceeds total hydrogen demand, avoiding under-sizing. Once the number of stations for each segment is estimated, the placement of individual HRS is done at equal intervals defined by the link/segment length. For a segment defined by the interior toll plaza i as the node, the HRS may be distributed at equal intervals within the segment. For each segment, the frequency of HRS is defined by:

$$Frequency\ of\ HRS = \frac{Link\ Length}{N_i^{total}}$$

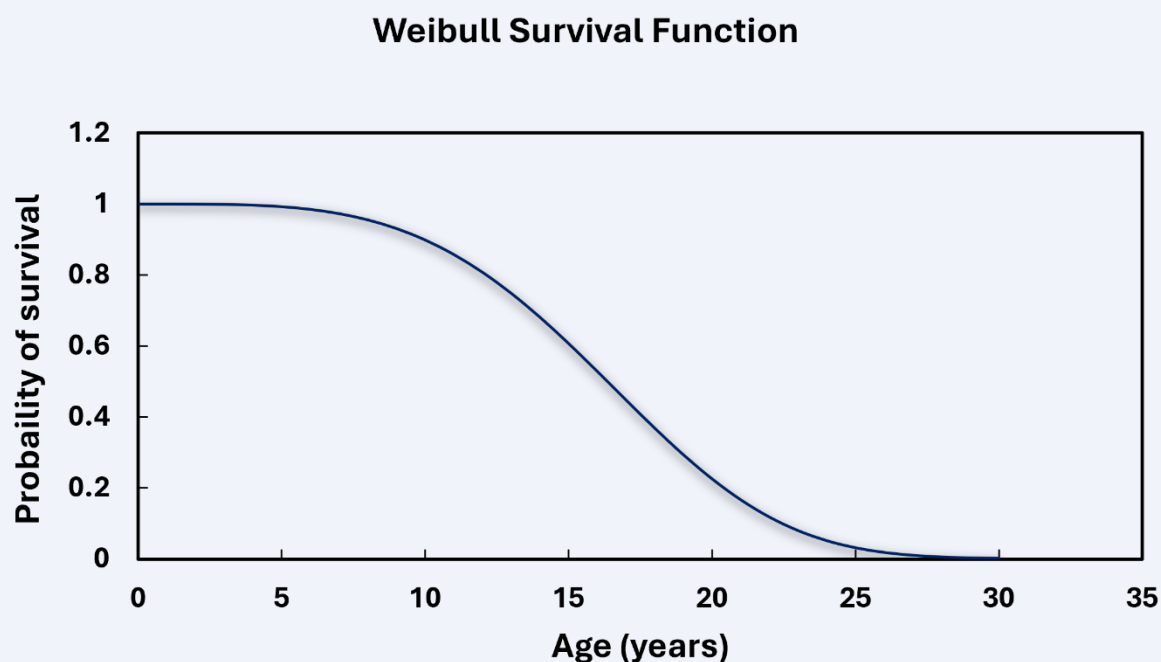
This discrete allocation approach reflects real-world deployment constraints:

- **Independency:** Each segment is considered as a separate and independent unit during HRS allocation. HRS requirements based on segment-wise hydrogen demand are met independently. This treatment also ensures that hydrogen FCEV trucks find HRS at multiple points within their travel range.
- **Demand matching:** Residual demand is satisfied using the smallest feasible station size, limiting overcapacity.

A.1.5 Fleet Projection

The active fleet of HCVs was estimated using a dynamic cohort-based stock-flow model, in which historical and projected vehicle sales are combined with age-dependent survival functions to reconstruct the in-use fleet over time. This model is effective in tracking how the fleet population evolves by grouping them in cohorts based on year. The total active fleet in any future year is calculated by applying a Weibull Survival Function to all historical and projected sales cohorts. This is widely applied in vehicle lifetime modelling and reliability analysis, representing the probability that a vehicle remains in operation as a function of age^{10,11}. This accounts for the gradual decommissioning of older vehicles due to mechanical wear and the 2021 National Vehicle Scrappage Policy. For the survival function, the shape, scale, and shift parameters are assumed to be 3.8, 18, and 0, respectively (under a high-survival-rate scenario for Indian HCVs). Cohort-based stock flow model accounts for the fact that an active fleet comprises the sum of the vehicles of each "age group" currently on the road.

Figure 2: Probabilistic Weibull survival function



A.2 Extended Results and Further Insights

A.2.1 Top 25 Segments with the Highest Annual HCV-VKT

Table 2: Top 25 segments with the highest annual VKT

Sl. No.	Segment Nodal Plaza	State	Major Towns/Section of Highway*	NH*	Total VKT (billion km)
1	Charoti	Maharashtra	Surat-Dahisar	8	1.34
2	Shahjahanpur	Rajasthan	Gurgaon-Kotputli-Jaipur Bypass	8	0.83
3	Nallur	Tamil Nadu	Chennai-Tada	16	0.78
4	Badauri	Uttar Pradesh	Chakeri-Usrania	2	0.77
5	Surajbari	Gujarat	Garamore-Samakhiali	27	0.72
6	Rasoiya Dhamna	Jharkhand	Barachetti-Gorhar	2	0.69
7	Lambiya Kalan	Rajasthan	Gulabpura-Chittorgarh Section	48	0.64
8	Nawabganj	Uttar Pradesh	Kanpur-Ayodhya	27	0.56
9	Mahuvan	Uttar Pradesh	Delhi-Agra	2	0.52
10	Kishangarh	Rajasthan	Jaipur-Kishangarh	8	0.51
11	Kalaparru	Andhra Pradesh	Vijayawada-Gundugolanu	5	0.48
12	Bhagwada	Gujarat	Surat-Dahisar	48	0.46
13	Veeravalli	Andhra Pradesh	Gundugolanu-Devarapalli-Kovvuru	16	0.46
14	Shrishikalan	Uttar Pradesh	Kabarai-Banda	76	0.45

15	Karapadu	Andhra Pradesh	Ichchapuram-Puintola	5	0.43
16	Baretha or Jajau	Uttar Pradesh	Agra-Gwalior	44	0.43
17	Sendurwafa	Maharashtra	Chhattisgarh/Maharashtra Border-Wainganga Bridge	6	0.42
18	Thakurtolla	Chhattisgarh	End of Durg Bypass- Chhattisgarh / Mah. Border	53	0.42
19	Shirpur	Maharashtra	MP/Maharashtra Border-Dhule	3	0.41
20	Krishnagiri	Tamil Nadu	Hosur-Krishnagiri	44	0.40
21	Bara Jode	Uttar Pradesh	Etawah-Chakeri	19	0.40
22	Laxmipuram	Andhra Pradesh	Ichchapuram-Narasannapeta	16	0.39
23	Ahmednagar Bypass	Maharashtra	Sinner-Shirdi-Ahmednagar	61	0.37
24	Saukala	Bihar	Aurangabad-Barachetti	19	0.37
25	L&T Panipat	Haryana	Panipat Elevated Highway	44	0.36

*Reference for NH numbers and section: (Indian Highways Management Company Ltd., 2025)⁸

A.2.2 Segments with a Hydrogen Demand of Over 100 TPD

Table 3: Segments with a hydrogen demand exceeding 100 TPD

Segment Nodal Plaza	State	Major Towns / Section of NH*	NH*	Hydrogen Demand (TPD)
Charoti	Maharashtra	Surat-Dahisar	8	330.3
Shahjahanpur	Rajasthan	Gurgaon-Kotputli-Jaipur Bypass	8	206.5
Nallur	Tamil Nadu	Chennai-Tada	16	192.4
Badauri	Uttar Pradesh	Chakeri-Usrania	2	189.6
Surajbari	Gujarat	Garamore-Samakhiali	27	178.3

Rasoia Dhamna	Jharkhand	Barachatti-Gorhar	2	171.0
Lambiya Kalan	Rajasthan	Gulabpura-Chittorgarh	48	157.3
Nawabganj	Uttar Pradesh	Kanpur-Ayodhya	27	138.3
Mahuvan	Uttar Pradesh	Delhi-Agra	2	127.9
Kishangarh	Rajasthan	Jaipur-Kishangarh	8	125.9
Kalaparru	Andhra Pradesh	Vijayawada-Gundugolanu	5	118.4
Bhagwada	Gujarat	Surat-Dahisar	48	114.4
Veeravalli	Andhra Pradesh	Gundugolanu-Devarapalli-Kovvuru	16	113.7
Shrishikalan	Uttar Pradesh	Kabara-Banda	76	111.0
Karapadu	Andhra Pradesh	Ichchapuram-Puintola	5	107.3
Baretha or Jajau	Uttar Pradesh	Agra-Gwalior	44	106.0
Sendurwafa	Maharashtra	Chhattisgarh/Maharashtra Border	6	103.4
Thakurtolla	Chhattisgarh	Chhattisgarh / Mah. Border	53	103.2
Shirpur	Maharashtra	MP/Maharashtra Border-Dhule	3	102.4
Krishnagiri	Tamil Nadu	Hosur-Krishnagiri	44	100.1
Bara Jode	Uttar Pradesh	Etawah-Chakeri	19	100.0

**Reference for NH numbers and section: (Indian Highways Management Company Ltd., 2025)⁸*

A.2.3 Class B (High Demand) Zones

The hydrogen demand, HRS estimates and distribution frequency for the high-demand segments for complete transition are summarised:

Table 4: Hydrogen demand and HRS estimates for Class B segments

Segment Nodal Plaza	State	Major Towns/Sections of NH*	NH*	Daily Hydrogen Demand (tonnes)	Total HRS	Frequency of HRS (km)
Shirpur	Maharashtra	MP/Maharashtra Border-Dhule	3	102.4	21	10.1
Karapadu	Andhra Pradesh	Ichchapuram-Puintola	5	107.3	22	14.1
Kalaparru	Andhra Pradesh	Vijayawada-Gundugolanu	5	118.4	24	11.6
Sendurwafa	Maharashtra	Chhattisgarh/Maharashtra Border-Wainganga Bridge	6	103.4	21	12.6
Laxmipuram	Andhra Pradesh	Ichchapuram-Narasannapeta	16	97.2	20	14.3
Veeravalli	Andhra Pradesh	Gundugolanu-Devarapalli-Kovvuru	16	113.7	23	14.0
Thakurtolla	Chhattisgarh	End of Durg Bypass-Chhattisgarh / Mah. Border	53	103.2	21	11.5
Shrishikalan	Uttar Pradesh	Kabrai-Banda	76	111.0	23	21.8
L&T Panipat	Haryana	Panipat Elevated Highway	44	89.2	18	10.0
Khawasa	Madhya Pradesh	Mohgaon to Khawasa	44	65.0	14	12.2
Kelapur	Maharashtra	Deodhari-Kelapur	44	46.0	10	13.3
Raikal	Telangana	Thondapali-Jedcherla	44	48.7	10	9.6
Raibha	Uttar Pradesh	Agra Bypass	3	88.1	18	10.1
Pagara	Madhya Pradesh	Guna Biaora Section	3	77.3	16	14.4
Chokkenahalli	Karnataka	Nelamangala-Tumkur	4	69.3	14	8.0
Mansar	Maharashtra	MP/Maharashtra Border	7	77.0	16	12.6
Manoharpura	Rajasthan	Gurgaon-Kotputli-Jaipur Bypass	8	51.6	11	5.9
Rajchandrapur	West Bengal	2 nd Vivekananda Bridge & Approach	12	58.4	12	7.6
Daffi	Uttar Pradesh	Varanasi-Aurangabad	19	54.2	11	5.2

Sasaram	Bihar	Varanasi-Aurangabad	19	46.8	10	7.0
Beliyad	Jharkhand	Barwa adda-Panagarh	19	70.5	15	10.6
Banskopa	West Bengal	Barwa adda-Panagarh	19	67.8	14	5.6
Palsit	West Bengal	Budbud-Palsit	19	71.9	15	4.8
Makhel	Gujarat	Samakhiyal-Santhalpur	27	70.8	15	0.9
Khemana	Gujarat	Palanpur/ Khemana- Abu road	27	52.0	11	10.0
Undvariya	Rajasthan	Abu road- Palanpur/Khemana	27	83.7	17	9.3
Ait	Uttar Pradesh	Poonch-Orai Section	27	88.3	18	10.4
Barhima	Bihar	UP-Border to Kotwa	27	47.2	10	13.9
Khedi	Rajasthan	Kishangarh to Gulabpura Section	48	59.4	12	8.2
Jojro Ka Kheda	Rajasthan	Gulabpura-Chittorgarh Section	48	75.1	16	6.7
Mandva (Narmada Bridge)	Gujarat	Vadodara-Surat	48	81.3	17	2.8
Choryasi	Gujarat	Bharuch-Surat	48	54.4	11	4.8
Boriach	Gujarat	Surat-Dahisar	48	61.3	13	4.4
Chennasamad uram	Tamil Nadu	Poonamalle- Walajahpet	48	23.3	14	4.3
Sriperumbadur	Tamil Nadu	Walajahpet- Poonamalai	48	14.6	11	2.9
Nagarhalla	Karnataka	Bijapur-Hungund	50	48.4	10	11.7
Pantangi	Telangana	Hyderabad-Vijayawada	65	53.5	11	9.9
Jamli	Madhya Pradesh	Khalghat- MP/Maharashtra Border	52	50.7	11	8.6
Mokha	Gujarat	Gandhidham (Kandla)- Mundra Port	41	47.4	10	4.6
Samakhiali	Gujarat	Samakhiyali- Gandhidham	41	75.0	15	3.1

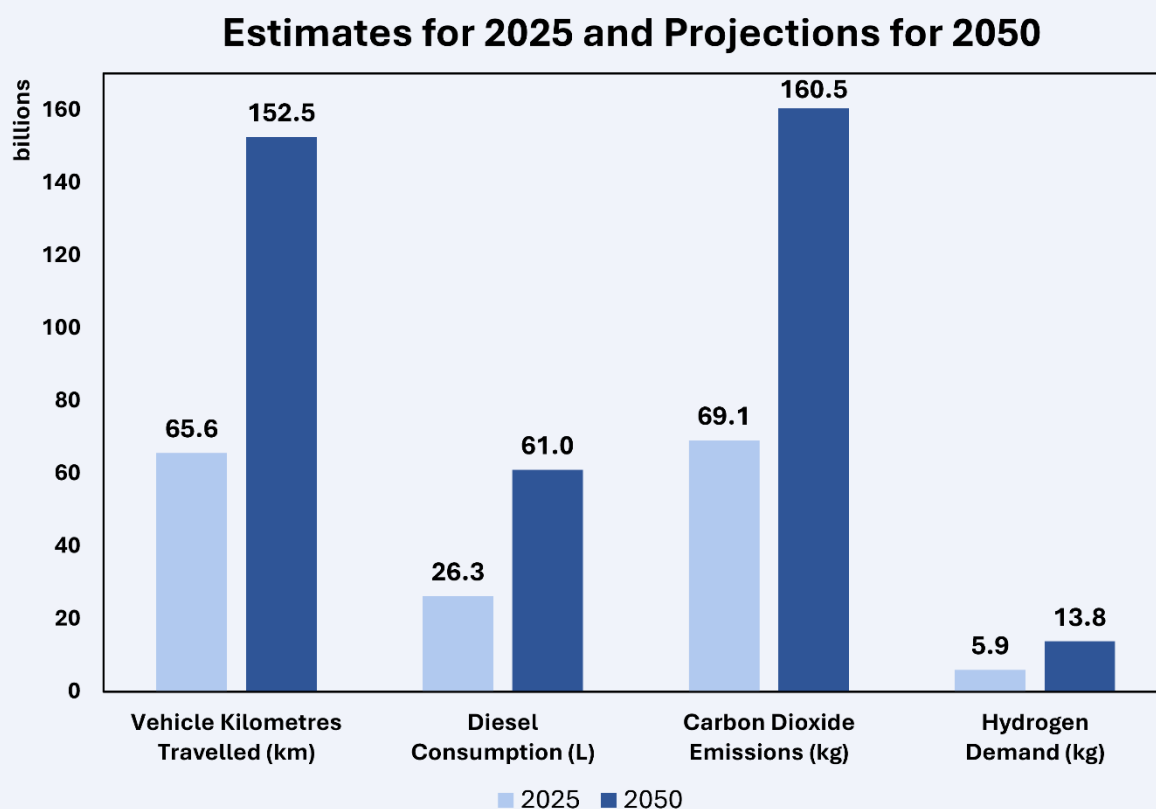
*Reference for NH numbers and section: (Indian Highways Management Company Ltd., 2025)⁸

Note: Frequency of HRS is defined here as the distance between successive HRS in km (when the HRS are distributed at equal intervals). It is determined by dividing the segment/link length by the total HRS in a segment.

A.2.5 HCV Fleet Projections for 2050

The active HCV fleet is estimated to be approximately 39,06,911 in 2025 and is expected to reach around 90,57,502 by 2050 (BAU scenario). The HCV-VKT is estimated to grow at an average of 3.43% yearly from 65.65 billion km in 2025 to **152.54 billion km** by 2050. Under the BAU scenario, the HCV diesel consumption on the highways will reach **61.02 billion litres** by 2050, contributing to annual CO₂ emissions of **160.47 Mt** by 2050. The annual hydrogen demand is expected to increase from 5.93 Mt to **13.8 Mt** by 2050.

Figure 3: HCV-specific VKT, diesel, carbon dioxide emissions, hydrogen demand estimates for 2025 and projections for 2050



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